Work-In-Progress: Software-Hardware Integration for Interactive Visualization of Mohr's Circle

Abstract

Mohr's Circle is a foundational tool in introductory Mechanics of Materials courses, used to visualize and analyze stress and strain in loaded structures. Despite its graphical nature, students often struggle to relate the abstract circle to the physical stress states it represents. To address this gap, an integrated learning system is being developed that pairs an instrumented aluminum tube specimen with a MATLAB application providing real-time visualization of stress states and Mohr's Circle. This Work-in-Progress paper reports on Phase II development, in which the system was advanced from a set of design specifications to a physically instrumented and experimentally validated prototype. Key contributions include the redesign of the measurement specimen from nylon to a 6061 aluminum tube, the replacement of an AD627-based analog signal chain with an HX711 24-bit integrated analog-to-digital converter (ADC), the placement and installation of two triaxial rosettes and two uniaxial gauges to isolate axial, bending, and torsional strain, and the development of a modular MATLAB application supporting both standalone user input and live serial data streaming. Experimental validation under axial and bending loads yielded calibration regression fits with R^2 values of 0.9993 and 0.9999, respectively, and the system achieved a peak-to-peak noise-limited strain resolution of 0.645 $\mu\varepsilon$ (6σ), well below the 1 $\mu\varepsilon$ design target. Future work includes torsional calibration and validation, prototype miniaturization, and a planned student effectiveness study.

Introduction

Mechanics of Materials (MoM) is a foundational undergraduate course that examines how stresses and strains develop in structural members under applied loading. Within this course, Mohr's Circle is introduced as a graphical tool for analyzing stress transformations [1]. While the method offers a compact way to determine principal stresses, maximum shear stress, and their orientations without repeated coordinate transformations, students consistently report difficulty connecting the abstract circle to the stress state of physically loaded structures [2], [3].

Prior teams [2], [3] established the conceptual foundation for an integrated learning tool that combines a publicly available MATLAB application for stress visualization with a handheld physical measurement device. The 2024 paper [2] introduced the MATLAB graphical user interface (GUI) and the initial SolidWorks design of the handheld tool. The 2025 paper [3] reported refinements to the app, fabrication of the mechanical components, and an outline of the planned electronic configuration, but explicitly identified strain gauge mounting, electronics installation, firmware development, and calibration as outstanding work.

This work-in-progress paper documents Phase II of the project, in which the system was advanced from a fabricated mechanical prototype with planned electronics into a physically instrumented and experimentally validated learning tool. The remainder of this paper outlines the educational motivation that continues to guide the project, summarizes updates to the mechanical, electrical, and software subsystems, presents experimental calibration and noise characterization results, and identifies remaining tasks for the next team.

Theoretical Background

The educational rationale for this learning system is grounded in experiential, embodied, and learning frameworks which contend the abstract formalisms are best acquired when paired with direct, sensorimotor experience [4]-[6]. Traditional MoM instruction relies heavily on a formalism-first approach, in which equations and analytical procedures precede physical experience [7]. While efficient for delivering content, this approach offers limited opportunities for students to develop intuition about the relationship between applied loads and the resulting internal stresses [8], [9].

Progressive formalization, or concreteness fading, has been shown to support transfer and retention by introducing concepts through concrete, perceivable referents before transitioning to idealized representations [10], [11]. Applied to Mohr's Circle, this means giving students the ability to physically apply axial, bending, and torsional loads to a structural member while simultaneously observing the resulting stress state on a digital Mohr's Circle in real time. The system reported here is designed to enable that pairing in an introductory MoM classroom setting.

Mechanical Design Updates

Specimen Selection

The structural specimen is a 1.5-inch outer diameter, 0.05-inch wall thickness 6061 aluminum tube. A circular cross-section was selected because its symmetric stress distribution under axial, bending, and torsional loading produces more predictable strain fields and more consistent measurements, which is critical for accurate stress-state visualization. Although other cross-sections may offer easier gauge attachment, the measurement accuracy and mechanical robustness of the circular geometry outweighed these drawbacks.

6061 aluminum was selected for its accessibility, machinability, and reliable elastic deformation under repeated loading. Compared to the nylon substrate specified by prior teams, aluminum offers superior surface preparation compatibility and bond reliability for strain gauge adhesion. Finite element analysis confirmed that the selected geometry produces measurable strain across all three loading modes defined in the design specification: 80 lbf axial compression, 10 lbf cantilever bending, and 30 lbf-in torque.

Gauge Placement and Installation

Load separation between axial, bending, and torsional measurements is achieved through gauge orientation and placement about the pipe. Rosette gauges (−45°, 0°, +45°) are mounted at two circumferential locations 180° apart, with the 0° element aligned to the pipe axis for axial strain and the ±45° elements capturing torsional shear strain. Two uniaxial gauges oriented parallel to the pipe axis are mounted 180° apart circumferentially, offset 90° from the rosette positions at the pipe midspan to maximize bending sensitivity. Each loading mode is measured with a dedicated Wheatstone bridge, a full bridge for torsion and a half bridge for axial and bending.

Electrical and Instrumentation Updates

Signal conditioning is handled by the HX711 24-bit ADC. An initial prototype using an AD627 amplifier with the ESP32's on board 12-bit ADC was found to be insufficient for this application.

Testing and Calibration

Noise-Limited Resolution

The noise-limited resolution of the system was quantified by analyzing the statistical variation in the measured strain signal produced by the tool under zero-load conditions, where any fluctuation in the signal can be attributed to electronic and sensor noise. A MATLAB program was used to collect 500 samples of raw counts from the microcontroller at a fixed sampling rate (10 Hz), and the standard deviation of this dataset was computed as a measure of the minimum resolvable signal. Preliminary results yielded a standard deviation of 516 counts, which represents the effective noise floor of the system.

The experimental standard deviation in counts was converted to an equivalent input-referred voltage noise using (1):

$$V_n = \frac{\sigma_N}{2^{23}} \cdot \frac{V_{ex}}{G} \tag{1}$$

Substituting measured values yields $V_n = 1.59\times10^{-6}$ V (1586 nV). This experimental noise floor exceeds the theoretical RMS noise floor of 33 nV by a factor of 50, suggesting external noise from the system dominates over the ADC noise floor..

After converting to strain units and applying a 20-sample simple moving average filter, the peak-to-peak noise-limited resolution, defined conservatively as 6σ of the filtered signal, is ε_{pp} = 0.645 με. This value remains within the 1 με target resolution defined in the design specifications. The raw and filtered signals are shown in Figure 1.

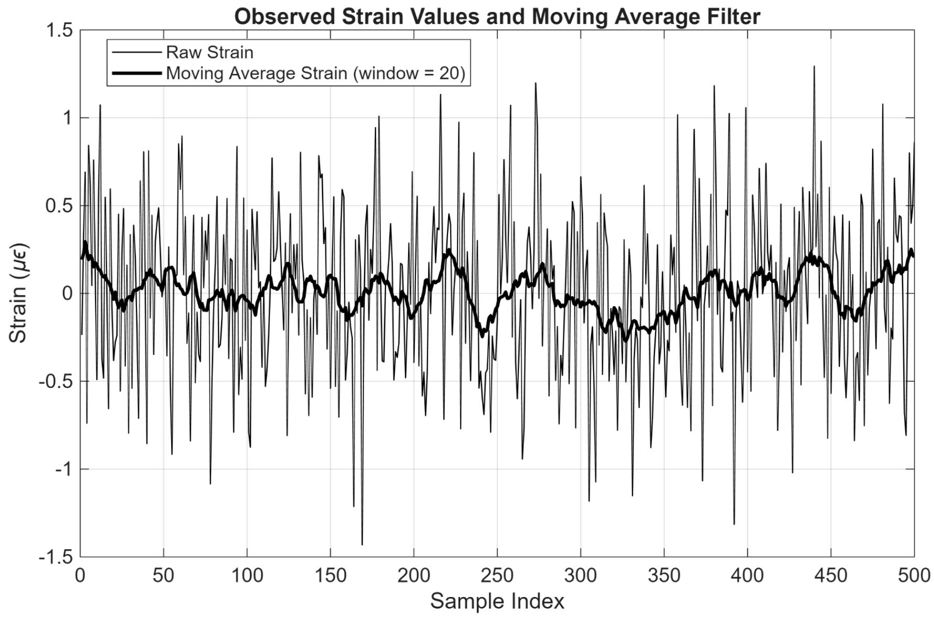

Figure 1. Raw strain signal and 20-sample moving average filter output over 500 samples, illustrating the noise reduction achieved by the SMA filter under zero-load conditions.

Axial and Bending Calibration

To determine the load calibration values for both axial and bending loading conditions, known loads were applied incrementally and the average sensor output was recorded at each load stage, with trials repeated across multiple runs to ensure repeatability.

Axial compressive loads were applied to the instrumented pipe in increments of 8 lbf, ranging from 0 to 80 lbf. Bending loads were applied to the pipe via cantilever loading, with a point load applied at the free end, ranging from 0 to 10.13 lbf using a series of seven progressively heavier calibration weights. At the start of each trial, a zero-load bias reference was recorded for offset correction.

After applying the zero-load bias adjustment for all axial and bending trials, linear regression was performed on the data to determine the calibration constant for each loading mode. The results of these linear regression studies can be seen in Figures 2 and 3, which confirm linearity between the applied load and sensor output.

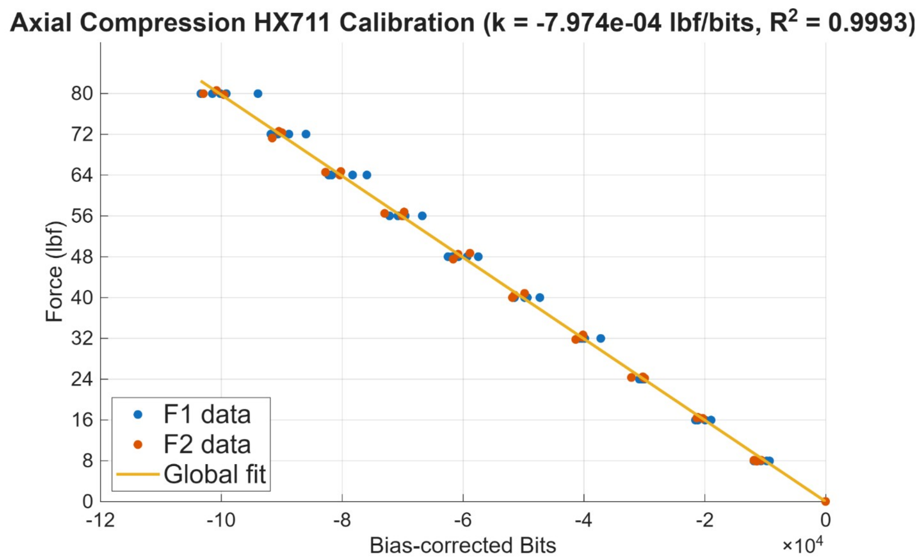

Figure 2. Axial compression HX711 calibration curves showing force versus bias-corrected bits for two datasets (F1, F2) with a global linear fit.

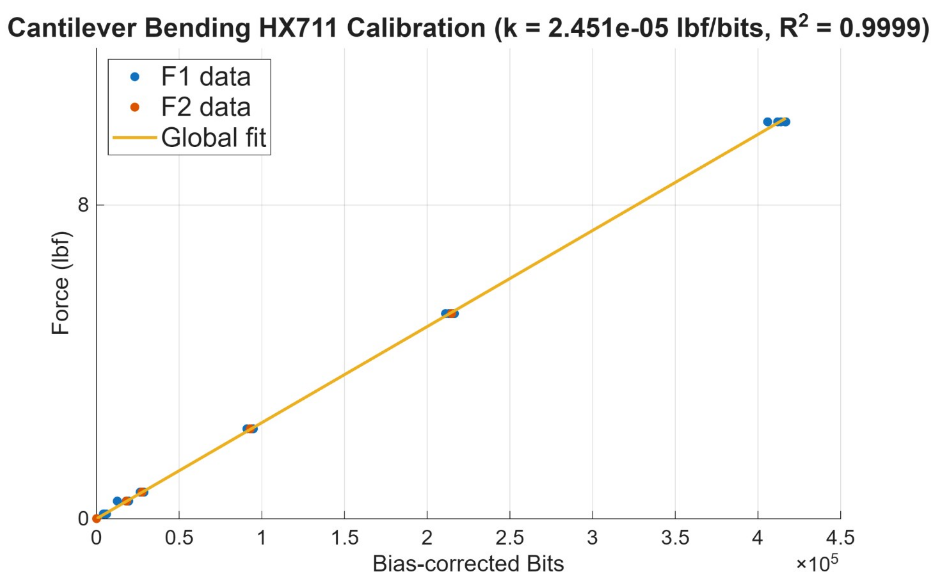

Figure 3. Cantilever bending HX711 calibration curves showing force versus bias-corrected bits for two datasets (F1, F2) with a global linear fit.

The theoretical force calibration constant for both axial and cantilever bending can be determined by evaluating (2) and (3):

$$k_P^{axial} = E \cdot A \cdot \frac{B}{G \cdot GF \cdot 2^{23}} \tag{2}$$

$$k_P^{bending} = \frac{E \cdot I}{(L-x)y} \cdot \frac{B}{G \cdot GF \cdot 2^{23}} \tag{3}$$

Substituting pipe geometry, material constants, and hardware configuration values yields calibration constants $k_P^{axial} = -1.92\times10^{-3}$ lbf/count and $k_P^{bending} = 6.473\times10^{-5}$ lbf/count. Comparison between the experimental and theoretical k_p values can be seen in Table 1.

Table 1. Comparison of theoretical and experimental calibration constants (k_p) for axial and bending loading cases.

Loading Type	Theoretical k_p (lbf/count)	Actual k_p (lbf/count)	Error (%)
Axial (compression)	-1.92×10^{-3}	-7.974×10^{-4}	-58.45
Bending (cantilever, end load)	6.473×10^{-5}	2.451×10^{-5}	-62.13

Future Work

Several tasks remain before the system is ready for classroom deployment. The most immediate hardware priority is torsional validation; the current scaled-up prototype geometry exceeded the size envelope of the available torsion test equipment, and a follow-up team will need to either re-scale the prototype or access alternative torsional fixtures to complete the three-axis calibration. The handheld housing must also be refined to integrate the HX711 modules, microcontroller, and wiring into a compact, durable enclosure suitable for repeated classroom handling.

On the software side, planned updates include further function decomposition for maintainability, the addition of in-app guidance windows for first-time users, coordinate-system overlays to reinforce the connection between physical loading and stress state, and colorblind-accessible color schemes for all plots. All stress computations should be systematically validated against benchmark textbook examples before classroom deployment.

Once hardware integration and software validation are complete, the next team will conduct the student effectiveness study described in [3], administering pre- and post-assessments in introductory MoM and Design of Machine Elements courses to evaluate whether the integrated system measurably improves student understanding of stress transformations and Mohr's Circle. The assessment instrument developed in [3] uses three levels of abstraction (less-abstract, more-abstract, and fully-abstract) mapped to four learning objectives, and a Bayesian linear regression model will be used to estimate effect sizes from the collected data.

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